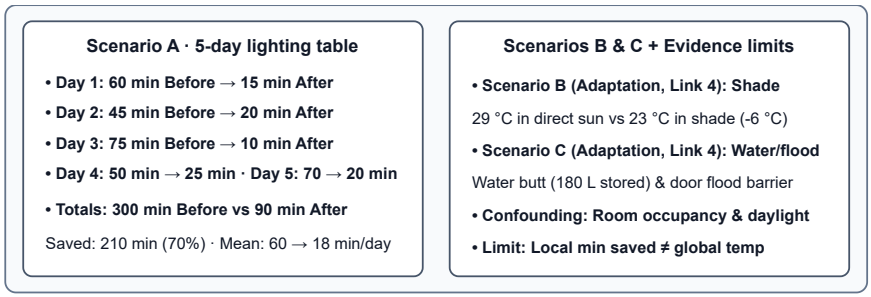
Reference sheets

Reference A · Safe Climate Action Scenarios (Scenarios A, B, and C) and 5-Day Practice Data

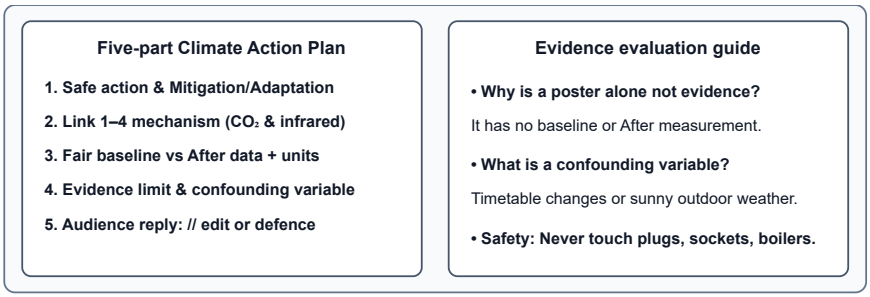


Five-day light table (300 min Before vs 90 min After). It also shows shade data (29 °C vs 23 °C) and data limits.

Reference only. Do not write on this sheet. All scenarios are safe paper datasets. Pupils never touch plugs, sockets, boilers, or live controls.

Scenario code.	Action and chain type.	5-day Before baseline vs 5-day After data.	Key Stage 3 mechanism and data limit.
Scenario A · Unused lights routine.	Turn off lights when room is empty. Mitigation (Links 1–3).	Days 1–5 Before: 60, 45, 75, 50, 70 min (300 min; 60 min/day). After: 15, 20, 10, 25, 20 min (90 min; 18 min/day). 210 min (70%) saved.	Cuts power demand (kWh), fossil fuel burning, and CO_2 infrared heat. Limit: measures room time, not kg CO_2 or global climate.
Scenario B · Shaded cooling space.	High-albedo light roof and shade in yard. Adaptation (Link 4).	5-day midday thermometer mean: 29 °C in direct sun. 23 °C under shaded roof (6 °C cooler).	Light roof (higher albedo) reflects sun rays and cuts heat at Link 4. Limit: does not cut CO_2 release at Link 1.
Scenario C · Water butt & flood gate.	Store roof rain and fit doorway flood gate. Adaptation (Link 4).	Water butt stores 180 L for dry days. Door gate keeps floor dry in a 25 mm rain test.	Cuts harm from drought and flood water at Link 4. Limit: does not remove greenhouse gases from the air.

Reference B · Five-part Climate Action Plan and fair-test guide



Five-part Climate Action Plan drawing. It shows safe action, mechanism, baseline vs after data, limits, and reply.

Plan part.	What to include in your plan.	Example from Scenario A (Unused lights routine).
1. Action & type.	Name the safe scenario. State Mitigation (Link 1) or Adaptation (Link 4).	'Scenario A (Unused lights routine) is mitigation at Link 1 and Link 2.'
2. Science mechanism.	Link the action to fossil fuels, CO_2, infrared heat, or Link 4 harm.	'Using less power cuts fossil fuel burning at power plants. That adds less CO_2 greenhouse gas to trap outgoing infrared heat.'
3. Baseline vs After data.	Compare Before baseline and After totals and means with units.	'Over 5 days, light waste fell from 300 minutes (60 min/day) Before to 90 minutes (18 min/day) After. That saved 210 minutes (70%).'
4. Limit & confounding factor.	State one confounding factor and what the data cannot prove.	'Confounding factor: room timetable changes or sunny weather. Limit: 5 days of room minutes cannot prove global climate cooling.'
5. Audience // reply.	Answer a review question and fix any over-claim with //.	'// Saved 210 minutes of wasted room light (Link 1 mitigation). It did not stop global warming in 5 days.'

Made by Matt

Tier pack · Aspire

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use Step 3 of the arrival drawing (5-day Before = 300 min; After = 90 min). Find the daily mean Before, the daily mean After, and the percentage drop.

Help. Divide each 5-day total by 5. Then divide the 210-minute saving by 300.

2. Wasted light time fell by 70%. Link this drop to the Key Stage 3 greenhouse gas mechanism (Links 1 to 3).

Help. Link kilowatt-hours, fossil fuel burning, carbon dioxide, and outgoing infrared heat.

3. Think about the 5-day Before baseline and 5-day After check. Why must they use the same room and lunch slot each day?

Help. Name what could make the test unfair if the room or time changed.

4. State one science limit of using empty-room light minutes as climate evidence.

Help. Tell local time saved apart from global CO_2 levels.

You do tasks

1. Use Reference A and Reference B. Compare Scenario A (Unused lights mitigation) and Scenario B (Shade adaptation and high-albedo roof). Link fossil fuel burning, carbon dioxide, greenhouse gas infrared heat, and surface albedo. Include the 5-day totals and daily means (300 to 90 min; 60 to 18 min/day; 70% drop). Staff I ☐ S ☐ C ☐

Help. Join the 5-day numbers with Key Stage 3 infrared and albedo ideas.

2. Check the fair-test rules of the 5-day light audit on Reference A. Name two confounding factors that could skew the After data. Explain how to control them. Staff I ☐ S ☐ C ☐

Help. Think about room use (timetables) and outdoor sun.

3. A listener asks: 'We saved 210 minutes of light in one room. Why can't we claim we cut air CO_2 by 70%?' Write a Key Stage 3 reply and // edit. **Staff I ☐ S ☐ C ☐**

Help. Tell a 70% drop in room light time apart from whole-air CO_2.

Optional participation-loop prompt. Share one part of your climate-action plan with an adult. Hear one review question or feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Sum up the numbers for Scenario A (Unused lights routine). Include 5-day totals, daily means, percentage change, and its Key Stage 3 mechanism.

2. How could a confounding factor like timetable shifts or sun skew the 5-day light audit? How do we control it?

3. Why must climate plans join Link 1 mitigation with Link 4 adaptation?

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Aspire

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt